Machine Learning Mini-Project Report

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Background

Multi-omics data has completely transformed cancer genomics as it allows researchers to characterise tumours at different molecular levels all at once.(1) One of these omics is RNA sequencing (RNA-seq) which evaluates gene expression, identifying which genes are active in a tumour.(2) Another is copy number alteration (CNA) data which captures DNA-level amplifications and deletions, ultimately providing complementary molecular views of the same tumour.(3) By integrating both of these data types, we are able to obtain a more complete picture of the tumour biology compared to either one of these omics alone, as a gene may be both amplified and overexpressed, strengthening confidence that the gene itself plays a functional role in disease progression. More specifically, genes that can be consistently identified across several machine learning methods should be of special interest, as cross-method agreement elevates confidence in genuine biomarker candidates.

For this project, the TCGA Breast Invasive Carcinoma dataset (PanCancer Atlas, n=1084) was used to study the basal-subtype breast cancer (Tripe-Negative Breast Cancer, TNBC).(4) The basal subtype was selected as it is known to carry the worst prognosis out of the major BRCA subtypes whilst also lacking targeted therapies and having the most distinct molecular profile.(5) These reasons meant that potential biomarker discovery was clinically significant. Based on this, the project focused on finding out whether multi-omics integration of RNA-seq and CNA data can identify survival-associated gene signatures in Basal subtype breast cancer.

To analyse this question, an elastic net penalised Cox regression for feature selection was first applied, before a Random Forest Survival model was utilised for variable importance validation. After that PROGENy pathway activity score was implemented and enrichment analysis was used to explain the context of identified genes with already known biological pathways. The mode robust biomarker candidates were then identified by looking at cross-method agreement between the elastic net Cox regression model and the survival Random Forest model.

Results

Result 1- Survival differs significantly across BRCA subtypes

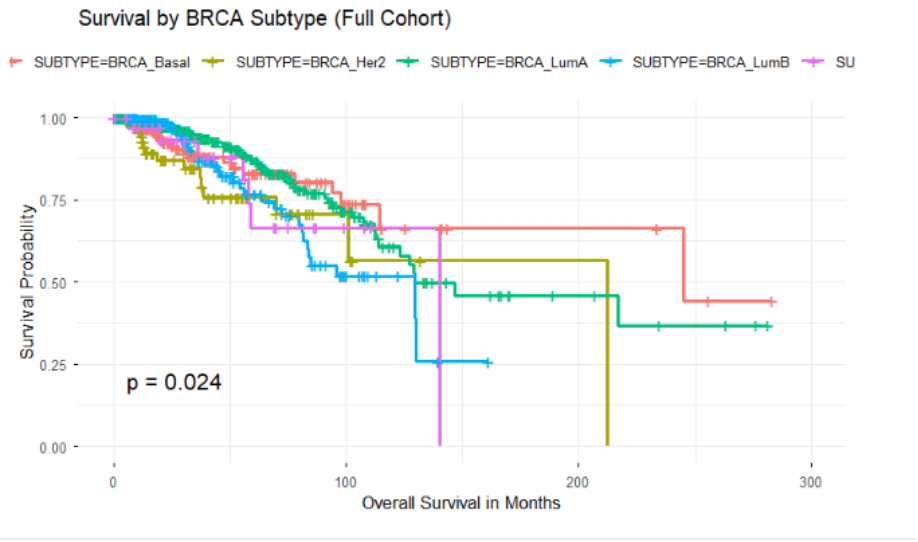


Figure 1: Kaplan-Meier overall survival curves across BRCA subtypes

Overall survival was compared across all subtypes in the full cohort (n=1084), to justify concentrating on the Basal subtype. As seen in Figure 1, the Kaplan-Meier curve (log-rank p=0.024) showed visible survival differences amongst BRCA subtypes, with large variations in survival trajectories between Her2, LumA, LumB and Basal patients. Although the different subtype curves overlap, the significant p-value of 0.024 confirms that that subtype is a suitable clinical variable. As noted in the Background section, analysis was centred around the Basal subtype due to its lack of targeted therapies, distinct molecular profile along with the clinical need to identify prognostic biomarkers in this difficult to treat subtype.

Result 2 – Basal Cohort Characterisation

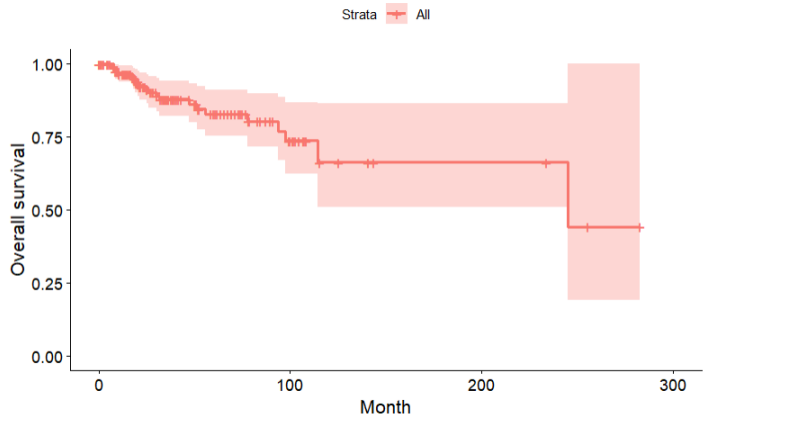


Figure 2: Kaplan-Meier overall survival curve for Basal patients

Following subtype filtering, a total of 171 Basal patients were identified. From Figure 2, it can be seen that the survival curve of this cohort illustrates a steady decline from 1.0 to around 0.45 over the course of 300 months, along with wide confidence intervals near the tail, which reflects the smaller sample size. It is also important to note that 22 death events were recorded across the cohort.

Result 3 – Multi-omics integration identifies amplified and over-expressed genes

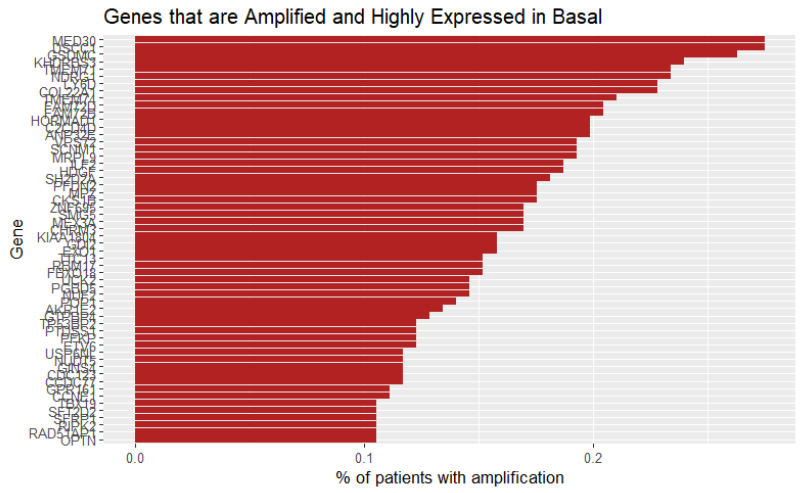


Figure 3: Genes that are both copy number amplified and highly expressed

By integrating the z-scores of the RNA-seq data with the CNA data across the 171 Basal patient cohort, genes that were both copy-number amplified (> 10% of patients) and highly expressed (mean z-score > 1) were identified. The CNA matrix obtained was comprised of 171 patients by 25,087 genes, whilst the combined multi-omics feature matrix consisted of 169 patients by 37,816 features. As illustrated in Figure 3, the most amplified and highly expressed genes included MED30, GSTMC and KHDRBS3. The finding that these identified genes were dysregulated at both the DNA and expression levels made them key candidates for biomarkers in Basal TNBC.

Result 4 – Pathway activity and correlations in Basal TNBC

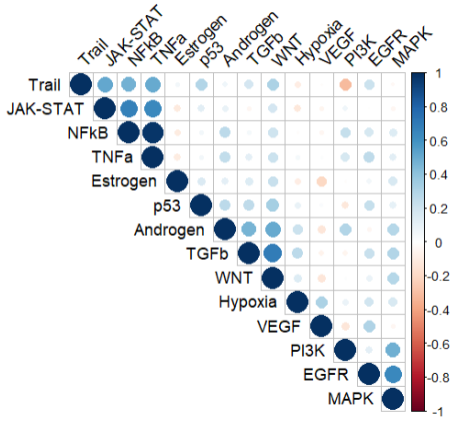


Figure 4: Correlation matrix of PROGENy activity scores across Basal patients

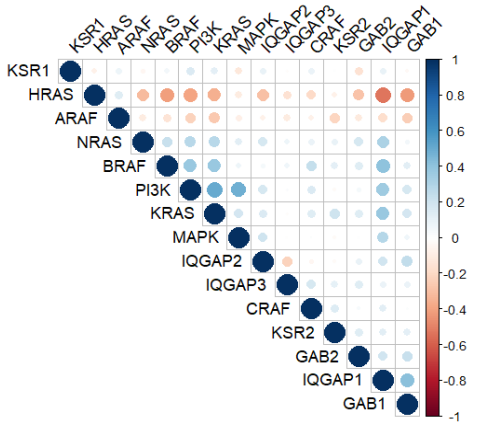


Figure 5: Correlation matrix of MAPK pathway gene expression scores

Through PROGENy pathway scoring, it was revealed that there was strong co-activation of inflammatory pathways in Basal TNBC. The pathway correlation plot seen in Figure 4 illustrated strong positive correlations between Trail, JAK-STAT, NFkB and TNFa signalling, which is in line with the inflammatory tumour microenvironment characteristic of TNBC.(6) The MAPK-focused correlation plot that can be seen in Figure 5 also highlighted strong correlation between BRAF, NRAS, KRAS and PIK3, which confirmed substantial co-activation of the RAS-RAF-MAPK axis.(7) Using the scores obtained by applying Glmnet over PROGENy, Hypoxia, MAPK, PI3K, Trail and VEGF were selected as the most survival-relevant pathways.

Result 5 – Elastic net feature selection identifies survival-associated genes

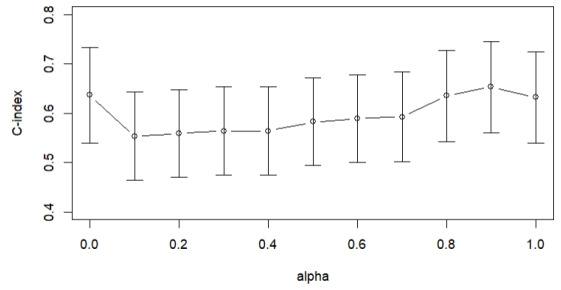


Figure 6: C-index across alpha values for RNA data only

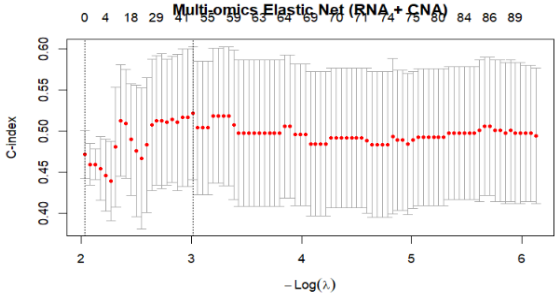


Figure 7: C-index across lambda values for multi-omics data

Table 1:Top survival associated genes identified by multi-omics elastic net Cox regression

GENE	COEFFICIENT	DIRECTION
SLC17A2	+1.333	Risk
PCNAP1	+1.023	Risk
KRTAP10-6	+0.706	Risk
ARHGAP36	+0.417	Risk
KRTAP19-1	+0.332	Risk
PDHA2	+0.395	Risk
OR10G7	-1.145	Protective
CELA2B	-1.031	Protective
IL9	-0.355	Protective
CETN1	-0.314	Protective
OSBPL6	-0.295	Protective
SLC18A1	-0.298	Protective

The combined RNA and CNA feature matrix (169 patients x 37816 features) was used for elastic net Cox regression (optimal α=0.9, selected by 10-fold cross-validation, favouring LASSO-like sparse selection). When looking at the RNA-only model, the alpha scan in Figure 6 illustrated a peak C-index of 0.65. This was considerably higher compared to the peak C-index of 0.52 that was achieved by the elastic net when applied to the combined multi-omics matrix, which also selected a mix of RNA and CNA features reaching a total of 45 features. Table 1 displays the top survival-associated genes, with positive coefficients indicating association with worse survival (risk genes) and negative coefficients indicating association with better survival (protective genes). As seen from Table 1 the top risk genes included SLC17A2, PCNAP1 and KRTAP10-6, whilst top protective genes included OR10G7, CELA2B and IL9. Additionally, CNA features such as CNA_EPM2A which showed the strongest positive coefficient and CNA_KMO which showed the strongest negative coefficient were also selected. This ultimately confirmed the value in integrating copy number data alongside gene expression. These top genes were then selected for the KEGG enrichment and STRING network analysis.

Result 6 – Cox Proportional Hazard Models

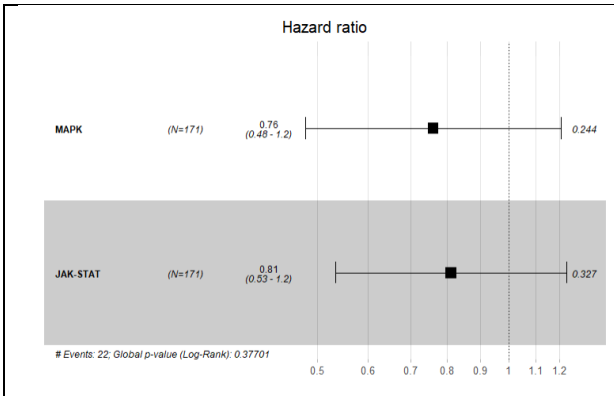


Figure 8: Hazard ratio plot for MAPK and JAK-STAT pathway activity scores in Basal patients

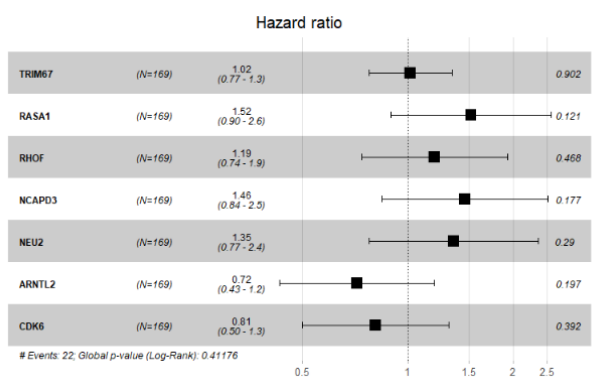


Figure 9: Hazard ratio plot for RF selected genes in Basal patients

Cox models were built for predictors at both the gene and pathway levels. The MAPK pathway seen in Figure 8 obtained a hazard ratio of 0.76 (95% CI: 0.48-1.2, p=0.244) whilst the JAK-STAT pathway in the same figure obtained a hazard ratio of 0.81(95% CI: 0.53-1.2, p=0.327), which meant that although both trends were protective, they did not achieve significance. This is likely down to the small number of event (n=22). In Figure 9 it can be seen that RASA1 had the greatest hazard ratio among the RF-selected genes, at 1.52(95% CI: 0.90-2.6, p=0.121) which suggested risk association. On the other hand, both CDK6 and ARNTL2 trended protectively, with hazard ratios of 0.81 and 0.72, respectively. With a global model p-value of 0.41, the limited power associated with only 22 events was once again highlighted.

Result 7 – MAPK pathway fails to significantly stratify Basal survival

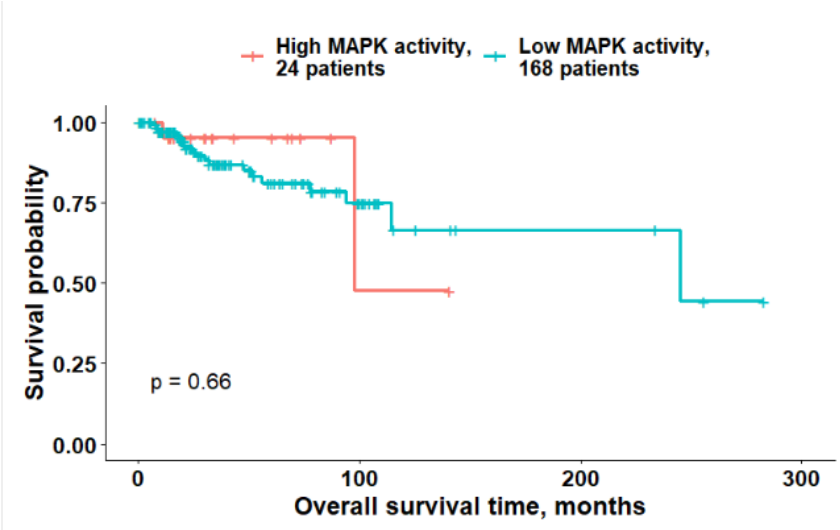


Figure 10: Kaplan-Meier overall survival curves for Basal patient stratified by MAPK pathway activity

Using an optimal threshold obtained by log-rank p-value scanning, patients were stratified into groups with high MAPK activity (n=24) and low MAPK activity (n=168). According to the Kaplan-Meier curve seen in Figure 10, there was no significant survival difference between groups (p=0.66), despite the high MAPK activity group appearing to maintain higher survival probability until around 90 months before dropping majorly. This result can be described as an informative negative result. The smaller size of the high MAPK activity group most likely hindered the comparison whilst MAPK activity alone appears to fail when trying to stratify Basal TNBC survival.

Result 8 – Random Forest validates key biomarkers identified

Table 2: Top 10 genes from survival Random Forest model ranked by VIMP score

GENE	APPROXIMATE VIMP
KRTAP10-6	0.008
FAM47A	0.007
KRTAP11-1	0.006
MTA2	0.0055
CASP8	0.005
SNORA50	0.0045
CPSF7	0.004
KCTD9	0.0038
IL1F10	0.0035
GPR68	0.003

On the training data, the survival Random Forest (RF) model (ntree=1000, importance=True) illustrated strong discriminative performance, obtaining a C-index of 0.985. According to the variable importance analysis (VIMP, scale= 0-0.008), summarised in Table 2, KRTAP10-6 was the strongest importance predictor of survival (~0.008). This was approximately 2.7 times greater than the tenth ranked gene GPR68(~0.003). Positive VIMP scores were displayed by all of the top ranked genes, which suggested that they actually contributed to survival prediction rather than just being noise. A interesting observation was that KRTAP10-6 also appeared as a top risk gene in the elastic net model, which clearly supports its role as a survival-associated biomarker across two independent analyses. The convergence of KRTAP10-6 across these two separate analyses, strengthens confidence in the fact that this gene could be a genuine biomarker in Basal breast cancer.

Result 9 – KEGG enrichment identifies metabolic and immune pathway involvement

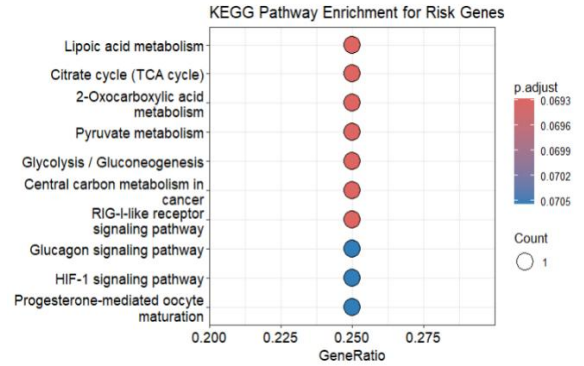


Figure 11: KEGG pathway enrichment plot for top risk genes

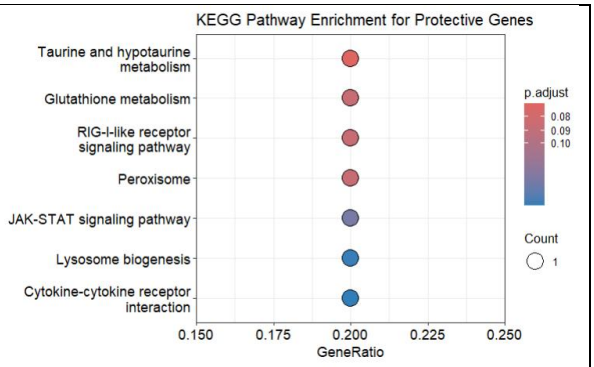


Figure 12: KEGG pathway enrichment plot for top protective genes

KEGG pathway enrichment with a pvalueCutoff of 0.2 and a qvalueCutoff of 0.5 was used on both the top risk and protective genes. Given the small number of genes actually submitted for the enrichment analysis stemming from the small Basal cohort size (n=171), an adjusted p-value cutoff of 0.2 was selected since the normal threshold of 0.05 returned no significant pathways. However, enrichment of top risk genes using this relaxed threshold identified significant pathways which included the TCA cycle, glycolysis/gluconeogenesis, pyruvate metabolism and HIF-1 signalling, indicating that poor survival in Basal TNBC is driven by metabolic reprogramming.(8) On the other hand, protective genes enriched in immune-related pathways such as JAK-STAT signalling, cytokine-cytokine receptor interaction and RIG-I-like receptor signalling, which is in line with the known immunogenic profile of TNBC.(9) One limitation that must be acknowledged is the fact that each pathway was represented by a single gene hit. This suggests that the small cohort size would benefit from validation in a much larger TNBC dataset.

Result 10 – STRING protein interaction networks

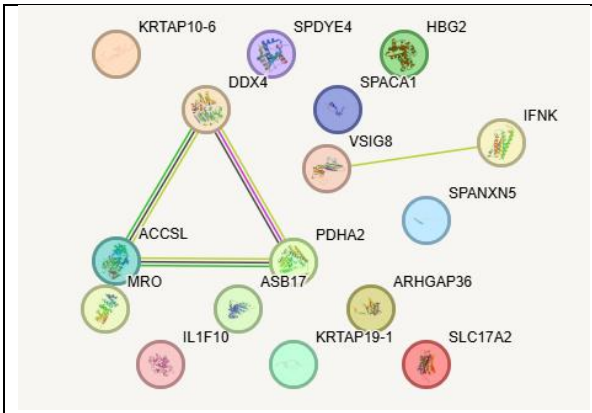


Figure 13: STRING protein interaction network for top 20 risk genes

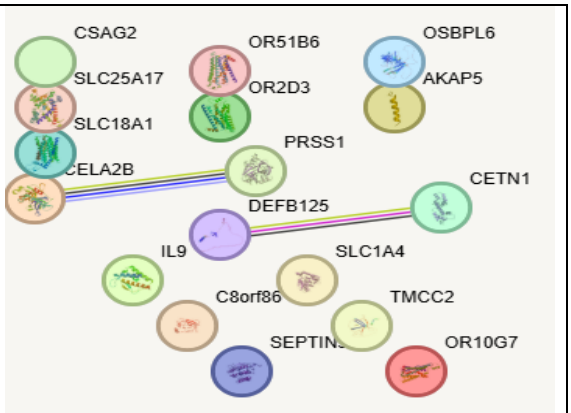


Figure 14: STRING protein interaction network for top 20 protective genes

As no interactions were detectable at the medium confidence threshold of 0.4, the separate STRING protein interaction networks as seen in Figures 13 and 14 were designed for the top 20 risk and protective genes at a more relaxed confidence threshold of 0.15 in the STRING database.(10) The failure to detect interactions at a higher threshold could indicate that some genes ae poorly characterised in existing databases. Figure 13, which represents the risk gene network illustrated a triangular interaction between ACCSL, MRO and ASB17 along with another interaction between VSIG8 and IFNK. In Figure 14, which represents the protective gene network, interactions between CELA2B and PRSS1 along with interactions between DEFB125 and CETN1 can be seen. These interactions further validate the immune and metabolic pathway enrichments found in the KEGG enrichment analysis. What is clear from these two networks and is somewhat consistent with the KEGG pathway enrichment analysis is the fact that the poor connectivity of both networks suggests that the survival associated genes found throughout this project actually represent a diverse signature rather than a co-regulated biological network.

Conclusion

Overall, the work done in this project demonstrated that integrating RNA-seq and CNA data in Basal breast cancer can identify a biologically relevant list of survival-associated genes. Some of the key findings of this project were the significant differences in survival across the various BRCA subtypes which justified the focus on the Basal subtype, the identification of genes such as MED30 and GSTMC which were both amplified and overexpressed at both DNA and RNA levels, and the elastic net identification of a multi-omics survival signature comprising of 45 features. After that, enrichment analysis identified immunological signalling pathways like JAK-STAT and cytokine-cytokine receptor interaction in protective genes, along with metabolic reprogramming pathways such as the TCA cycle and glycolysis in risk genes, both of which are consistent with the known biology of TNBC.(11)

As MAPK pathway activity alone failed to significantly stratify Basal survival ($p=0.66$), it suggested that single-pathway stratification is insufficient for this heterogenous subtype and that instead, multi-omics gene signatures are necessary. This reiterates the most compelling finding of this analysis even further, which was the convergence of KRTAP10-6 as a top survival-associated gene across two independent approaches, elastic net Cox regression (coefficient=0.706) and survival Random Forest (highest VIMP score). This finding provides a strong argument for KRTAP10-6's candidacy as a prognostic biomarker in TNBC. With that being saying, it is also important to highlight some of the limitations of this analysis. One being that the small number of death events (n=22) reduced statistical power, with another being that the limited cohort size was reflected in the single-gene KEGG pathways hits. The final limitation of the project was the likely overfitting of survival Random Forest model on the training data observed by the C-index score of 0.985. Therefore, I think that future work should be focused on validating the KRTAP10-6 survival association using an external, independent TNBC cohort.

References

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